

METHYLDOPA [ALDOMET]

Pharma Talk

Antihypertensive + Central Alpha 2 Agonist

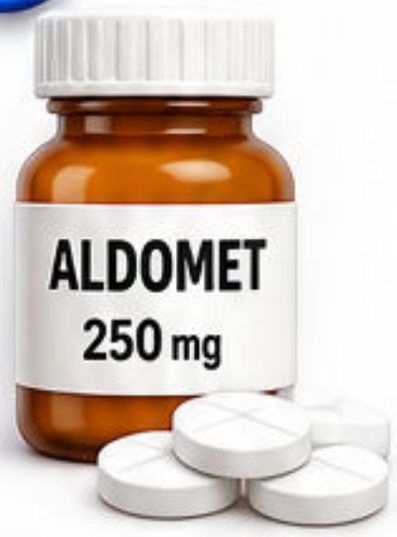
A centrally acting antihypertensive that reduces sympathetic outflow to lower blood pressure.

1 MECHANISM OF ACTION (MOA) & TARGET

- **TARGET:** Central alpha 2 adrenergic receptors.
- **MOA:** Prodrug → alpha-methylnorepinephrine → stimulates central alpha 2 brainstem receptors → ↓ sympathetic outflow → ↓ systemic vascular resistance and ↓ BP.
- **Analogy:** A central brake pedal for the nervous system.



2 FORMS & STRENGTHS



Tablets:
250 mg, 500 mg

Oral Suspension:
250 mg/5 mL

3 INDICATIONS (FDA-APPROVED & MAJOR OFF-LABEL)

FDA APPROVED:

- Hypertension

MAJOR OFF-LABEL:

- Gestational hypertension (historical use)
- Hypertensive emergency in pregnancy

4 DOSAGE & ADMINISTRATION

A) ADULT DOSE

- **Standard:** 250 mg two to three times daily
- Titrate every 2 days
- **Max:** 3 g daily
- **Route:** PO
- With or without food

B) PEDIATRIC DOSE

- **All ages** (Infant–Adolescent)
- **Dose:** 10 mg/kg daily divided
- **Route:** PO
- **Max:** N/A
- Adjust for renal function

C) ADMINISTRATION



- Route: Oral
- With or without food
- Swallow tablets whole
- Shake suspension well before use

D) ADJUSTMENT



- **Renal:** CrCl <50 mL/min → decrease frequency
- **Hepatic:** Avoid in hepatic impairment

5 PHARMACOKINETICS & DYNAMICS (PK/PD)



- **Onset:** 4–6 hours
- **Elimination $t_{1/2}$:** 1.7 hours
- **Metabolism:** Hepatic (prodrug)
- **Excretion:** Renal

6 ADVERSE EFFECTS & TOXICITY

COMMON:

- Sedation
- Dry mouth
- Headache
- Depression
- Orthostatic hypotension
- Weight gain

SEVERE (RARE):

- Hemolytic anemia
- Hepatotoxicity
- Positive Coombs test
- Drug induced lupus



7 CONTRAINDICATIONS & INTERACTIONS

ABSOLUTE CONTRAINDICATIONS:



- Active hepatic disease
- Monoamine oxidase inhibitor (MAOI) use

MAJOR DRUG INTERACTIONS:



- MAOIs → ↑ catecholamines → hypertensive crisis
- Iron → ↓ absorption
- Additive sedation with CNS depressants

8 SPECIAL POPULATIONS (PREGNANCY & LACTATION)

PREGNANCY	LACTATION	GERIATRIC	PEDIATRIC
 Safe Category B	 Excreted in milk; use caution	 ↑ Risk of sedation and orthostasis	 Dose by weight; monitor closely

9 CLINICAL PEARLS



- Causes false positive urine catecholamines.
- Check baseline CBC and LFTs.
- ~20% develop positive direct Coombs test; rarely causes actual hemolytic anemia.
- Rebound hypertension if stopped abruptly.
- Labetalol or nifedipine preferred first line agents in pregnancy.

10 GLOSSARY OF ABBREVIATIONS



BP	=	Blood pressure
CBC	=	Complete blood count
CrCl	=	Creatinine clearance
LFTs	=	Liver function tests
MAOI	=	Monoamine oxidase inhibitor
PO	=	By mouth (oral)
PK/PD	=	Pharmacokinetics/Pharmacodynamics
Susp	=	Suspension